ADRIKA PRADHAN
M.Sc. Artificial Intelligence, USI Lugano (Switzerland) | adrika.pradhan@usi.ch |
+41 767516652 | LinkedIn: https://www.linkedin.com/in/adrika-pradhan-3a92622ba/ |
Github:https://github.com/adriennie | Indian national, Swiss student permit (B)

PROFILE SUMMARY

M.Sc. Artificial Intelligence student at USI Lugano (B.Tech Computer Science, AI & ML). Experience in evaluating deep-learning pipelines on noisy scientific data (astronomical image denoising + YOLO-style detection) and in benchmarking LLM inference and fine-tuning on limited GPU resources (vLLM, PEFT/LoRA). Seeking a Technical Studentship to apply and extend these skills in a supervised, team-based engineering setting.

CORE SKILLS

- Languages: Python, SQL, TypeScript, Java
- ML / AI: PyTorch, Keras/TensorFlow, scikit-learn, LightGBM, Hugging Face Transformers/TRL, PEFT/LoRA, vLLM,
- Methods: CNNs and autoencoders, YOLO-style detection, anomaly detection (Isolation Forest), RAG evaluation
- Evaluation: PSNR, SSIM, IoU, throughput and latency benchmarking
- Data & systems: PostgreSQL/PostGIS, Supabase (row-level security), REST APIs, Google Cloud Run, Git
- Mathematics: linear algebra, probability and statistics, discrete mathematics

SELECTED PROJECTS

vLLM PagedAttention vs. Hugging Face Transformers (Benchmark : Python, PyTorch, vLLM, Tesla T4)
Link: https://github.com/adriennie/vllm-pagedattention-benchmark

- Benchmarked vLLM (PagedAttention) against Hugging Face generate() on Qwen2.5-1.5B-Instruct for batch sizes 1-32 on a Tesla T4; at batch 32, vLLM reached 877.69 tok/s versus 173.45 tok/s (5.06x), or 1.43x against a second Hugging Face run in the same data.
- Documented the limits of the measurements (runtime versions not pinned, vLLM memory not captured) and planned device-level memory profiling.
- Reproducible Colab/Kaggle notebooks, CSV results and plots in a public repository.

LoRA Rank Ablation & Multi-Adapter Serving with vLLM (Python, PEFT, TRL, vLLM)
Link: https://github.com/adriennie/vllm-lora-extension

- Trained LoRA adapters (q_proj, v_proj; ranks 4, 8, 16) on Qwen2.5-1.5B-Instruct on a Kaggle T4; trainable parameters grew from 0.54M to 2.18M (0.035% to 0.14%).
- Served all three adapters from one vLLM engine using LoRARequest routing.
- Documented limits (four unique training examples, training loss only, no held-out evaluation) and planned seeds, a held-out set and a serving benchmark.

Swasthya:Intelligent Pharmacy Network (Multi-Role Pharmacy Network Platform)
Link: https://github.com/adriennie/swasth-app

- Built a multi-role (admin, distributor, pharmacy, customer) supply-chain app with a relational schema, row-level security and a REST API.
- Deployed a LightGBM demand-forecasting service and an Isolation Forest anomaly-detection pipeline as separately versioned backend services.

EcoShare (Full-Stack Community Credit Platform)
Link: https://github.com/adriennie/Revive-App

- Implemented atomic credit transfers, an escrow engine and expiry detection with per-user isolation through row-level security.
- Used PostGIS for proximity ranking and spatial clustering; built an NGO auto-matching algorithm and real-time notifications.

RESEARCH

Research Project – Denoising Astronomical Images with Convolutional Autoencoders to Improve YOLO-Based Object Detection
Link: https://github.com/adriennie/Astronomical-Image-Processing

- Trained a convolutional autoencoder from scratch on DeepSpaceYOLO images (128x128, synthetic Gaussian noise, σ=0.2) and a YOLO-inspired 16x16-grid detector; compared detection on noisy vs. denoised inputs.
- Test-set PSNR 27.52 dB, SSIM 0.88 (highest SSIM vs. DnCNN, U-Net, YOLOv1 and a standard autoencoder; PSNR second to DnCNN and U-Net). Mean IoU 0.51 → 0.69 on denoised inputs, measured against placeholder labels.
- Manuscript written in LaTeX; code and notebook in a public repository.

EXPERIENCE

Generative AI Engineering Intern – Google Cloud Track NASSCOM FutureSkills | Remote |
NASSCOM | *Jun '25 — Jul '25*

- Built and evaluated retrieval-augmented generation (RAG) pipeline prototypes using Gemini models and Vertex AI, focusing on production-readiness and reproducible configurations
- Designed structured evaluation frameworks for measuring output faithfulness and multimodal document processing quality
- Delivered documented pipeline specifications suitable for handoff and deployment

---

EDUCATION

M.Sc. Artificial Intelligence, Università della Svizzera italiana (USI), Lugano | Sep 2026 – expected August 2028
Current courses: Numerical Algorithms, Engineering AI Systems, Deep Learning, Machine Learning, Algorithm and Complexity
B.Tech Computer Science (AI & ML), UPES, India | 2022 – 2026
CGPA 7.91/10 | Final two semesters: 9.18 and 10.0 SGPA

CERTIFICATES

- Microsoft Certified: Azure AI Engineer Associate (AI-102) · April 2026 · Credential: P8376A-B25D52
- AWS: Building Large Language Models on Amazon Bedrock
- Python & Statistics — HKUST

LANGUAGES

- English: professional working proficiency (IELTS 6.5, CEFR B2)
- German — A1 (in progress)

ACHIEVEMENTS

- TCS CodeVita 2025 — Advanced to Round 2 (Rank: 2083 / 6138)
- Code4Bharat Hackathon — National Semifinalist.
- Care-a-Thon (Sukhyam-sponsored) — Finalist; awarded ₹50,000 innovation grant